

1. Customer orders

- High-mix, low-volume
- Different part geometry
- Heterogeneous due dates

Order 1 (due date d_1)



Order 2 (due date d_2)



Order 3 (due date d_3)

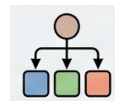


...

Order n (due date d_n)



2. Scheduling decisions



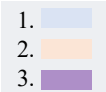
Part-to-machine assignment

Assign each part to one of the parallel AM machines.



Batch formation

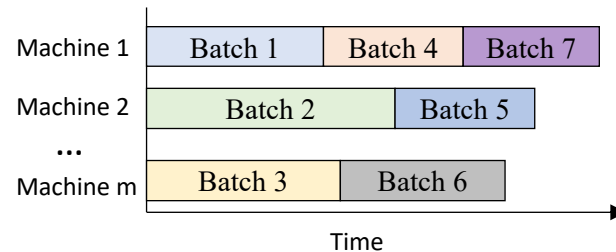
Group assigned parts into batches under the build capacity.



Batch sequencing

Determine the processing order of batches on each machine.

Output: Production schedule



3. Parallel AM machines



AM machine 1



Batch 1



Batch 4



Batch 7



AM machine 2



Batch 2



Batch 5



AM machine 3



Batch 3



Batch 6



Batch processing

The batch processing time depends on the maximum part height and the total volume of parts in the batch.

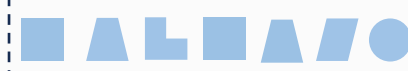
4. Part completion and order delivery



Part availability for delivery

Parts in a batch complete simultaneously, whereas an order is deliverable only after all its parts are completed.

Order 1



Order 2



Order 3



Order 4



Order submission



Order delivery



Customers